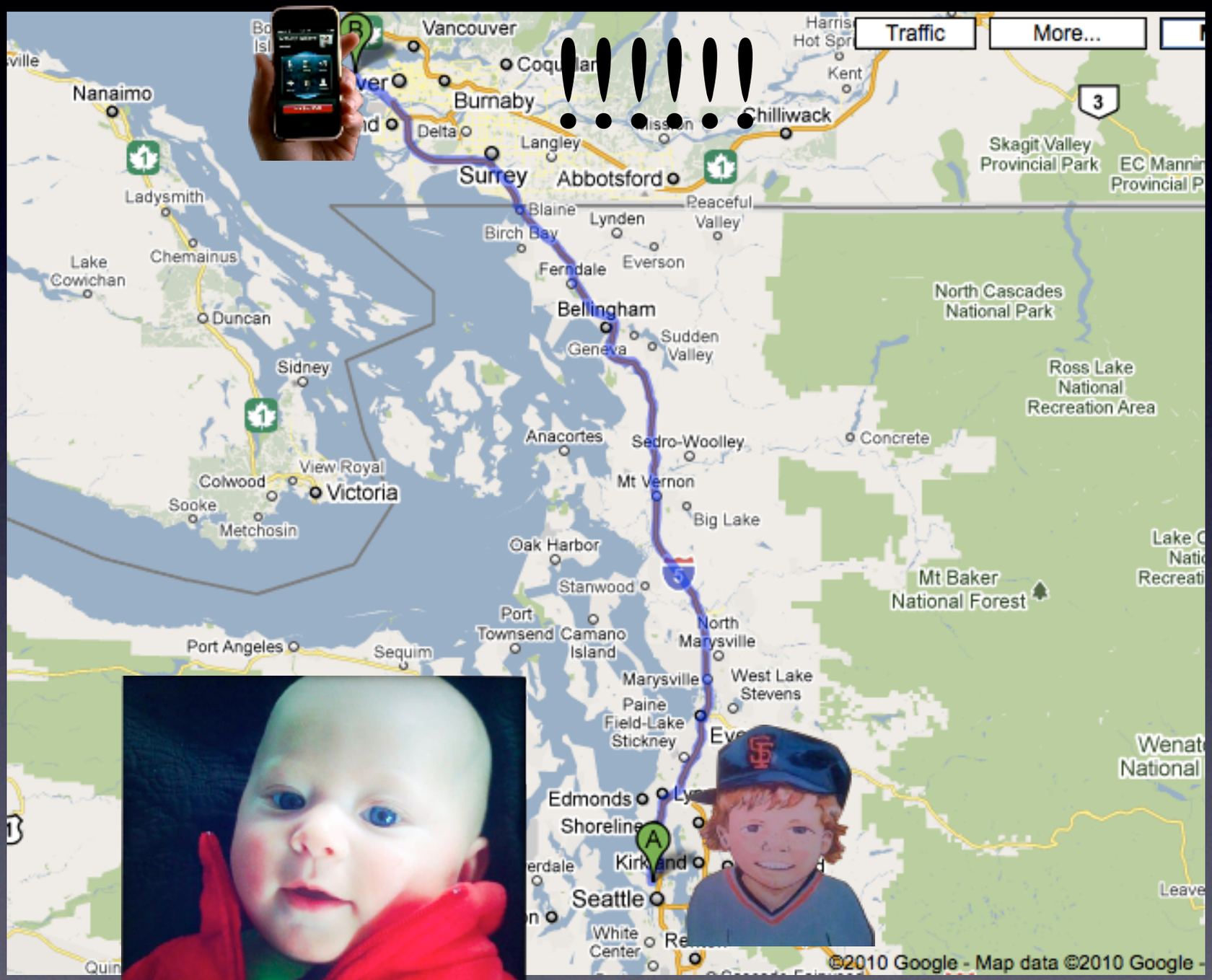
Help! There's a Computer in My Physical System!



Dave Bacon
Department of Computer Science & Engineering
University of Washington

Oh Canada!


!!!!!!
Traffic
More...
Vancouver
Burnaby
Surrey
Delta
Langley
Abbotsford
Chilliwack
Kent
Nanaimo
Ladysmith
Chemainus
Lake Cowichan
Duncan
Sidney
Colwood
View Royal
Victoria
Sooke
Metchosin
Blaine
Birch Bay
Lynden
Peaceful Valley
Everson
Ferndale
Bellingham
Sudden Valley
Geneva
Anacortes
Sedro-Woolley
Mt Vernon
Big Lake
Oak Harbor
Stanwood
Port Townsend
Camano Island
North Marysville
Marysville
West Lake Stevens
Paine Field-Lake Stickney
Port Angeles
Sequim
Edmonds
Shoreline
Kirkland
Seattle
White Center
Skagit Valley Provincial Park
North Cascades National Park
Ross Lake National Recreation Area
Mt Baker National Forest
Concrete
©2010 Google - Map data ©2010 Google

Quantum Computers

Condensed

1994

1995

1995

2002

2005

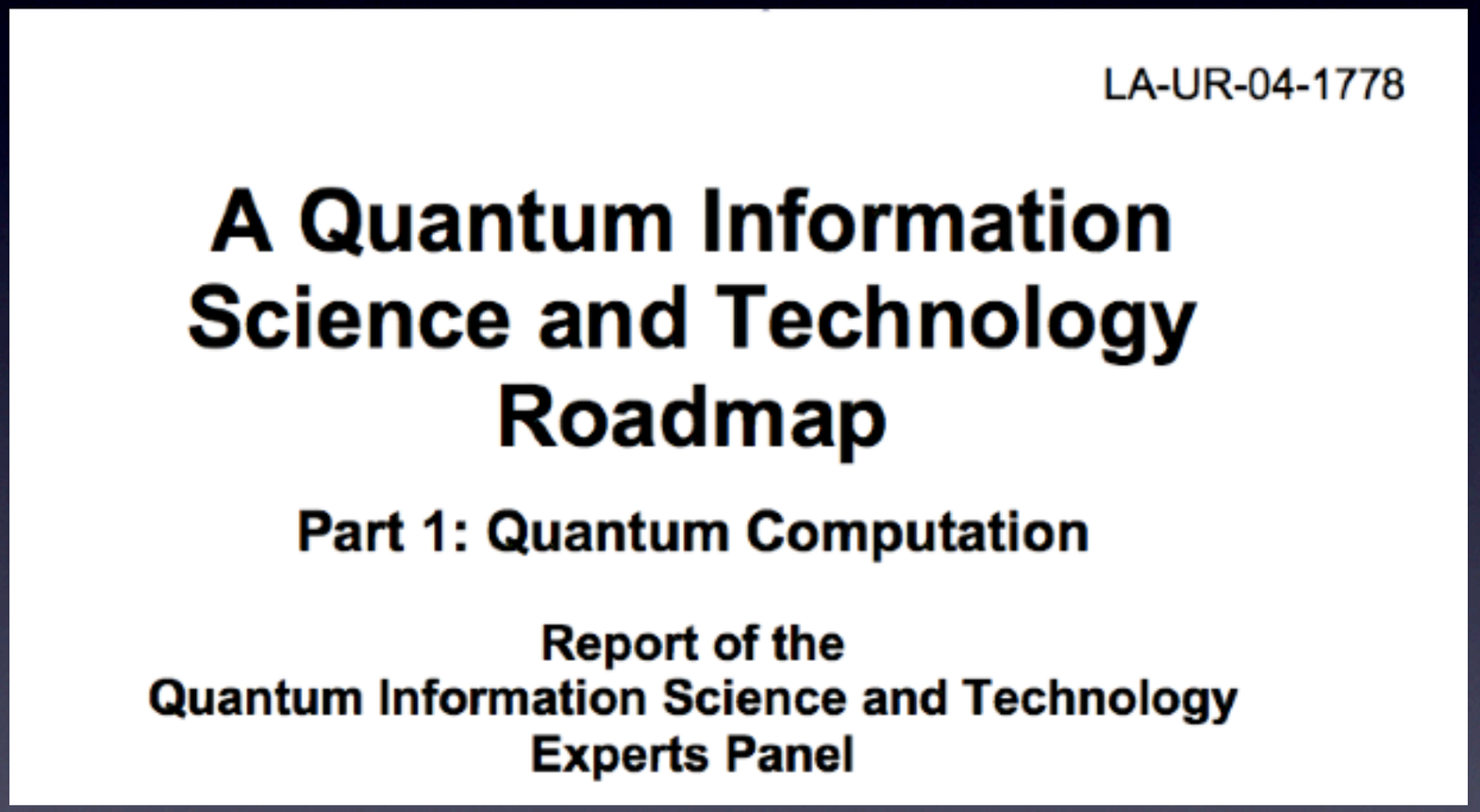
2006

Peter Shor shows that a quantum computer could

Why Don't We Have a Quantum Computer Yet?

A Plan![*]

[2002 and 2004 roadmap:]



LA-UR-04-1778

A Quantum Information Science and Technology Roadmap

Part 1: Quantum Computation

Report of the
Quantum Information Science and Technology
Experts Panel

http://qist.lanl.gov/qcomp_map.shtml

[*]but no man, canal, or Panama

Table 4.0-1

The Mid-Level Quantum Computation Roadmap: Promise Criteria

QC Approach	The DiVincenzo Criteria: Quantum Computation #1	#2	#3	#4	#5	QC Networkability #6	#7
NMR							
Trapped Ion							
Neutral Atom							
Cavity QED							
Optical							
Solid State							
Superconducting							
Unique Qubits	This field is so diverse that it is not feasible to label the criteria with "Promise" symbols.						

1. "scalable" system of qubits
2. qubit initialization
3. long decoherence times
4. universal set of quantum gates
5. qubit measurement

achieved
proposed
no viable method known

The Roadmap



Where experimental QC dedicates most of its resources

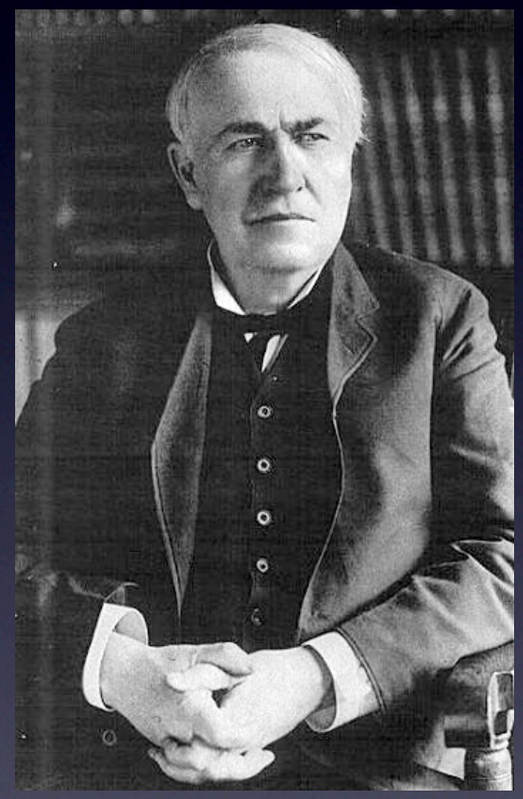
Threshold theorem

~ Given quantum computer components which fail with probability less than a threshold, one can combine these to make a robust quantum computer

Roadmaps are great for getting funding,
but a horrible way to develop a technology?

Why Don't We Have a Quantum Computer Yet?

"To invent, you need a good imagination, and a pile of junk"



Thomas Edison

Give Me A Junkyard ;)

Where are the teams trying to build hundreds, thousands, millions, billions of “bad qubits”?

D:wave

The Quantum Device Company™

Imagination

"To invent, you need a good imagination, and a pile of junk"
- Thomas Edison

Why should we trust that our imagination can make use of a pile of junk?

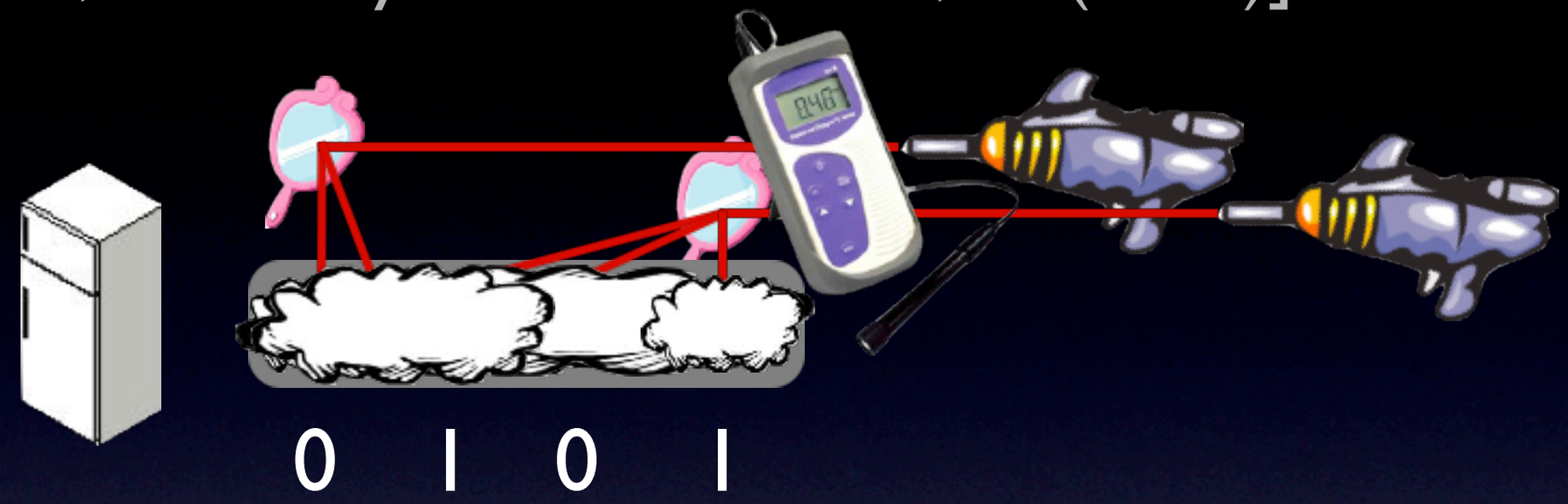
Models of Quantum Computing

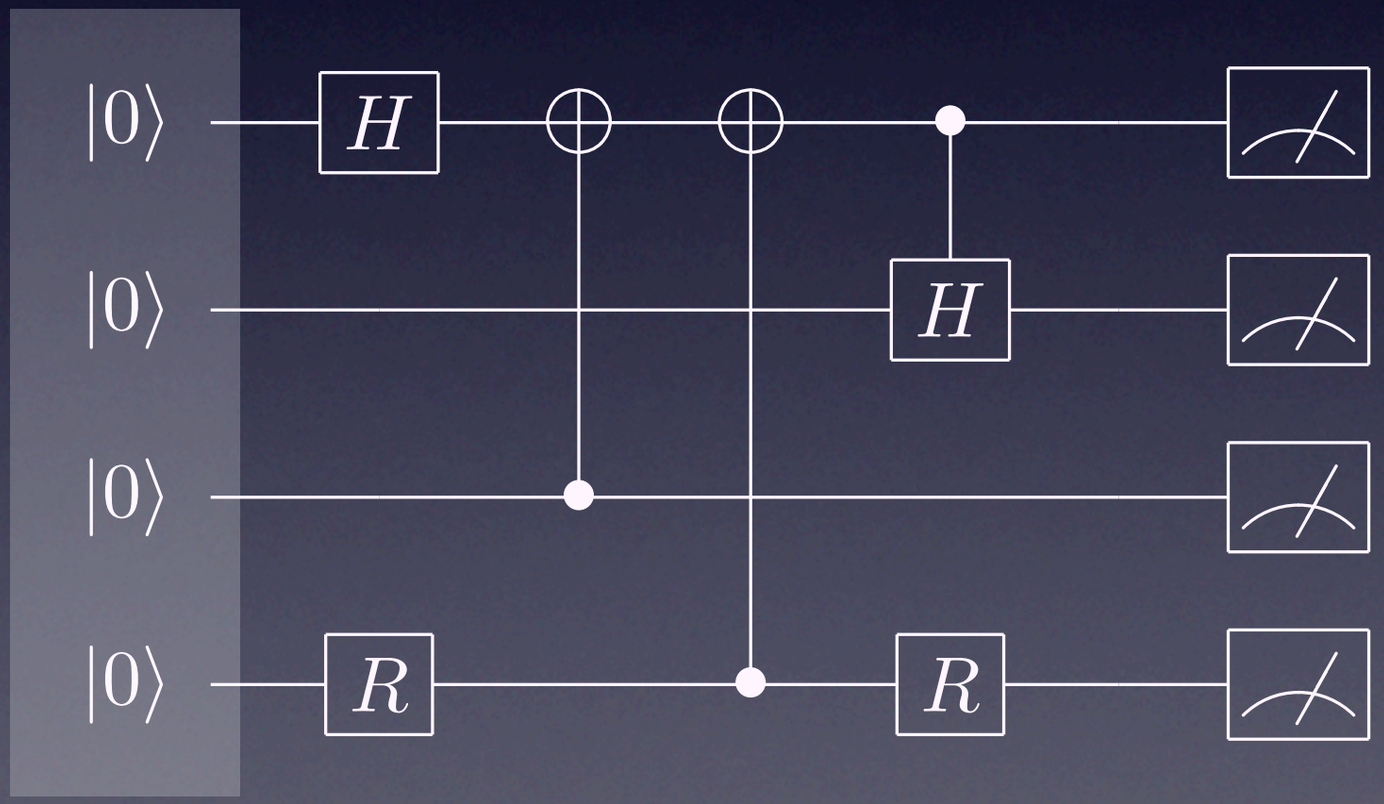
Models of QC

- Quantum Circuit Model
- Measurement Based Quantum Computing
  - One-way model
  - Teleportation model
- Universal Adiabatic Quantum Computing
- Topological Quantum Computing
- Holonomic Quantum Computing
- Quantum Random Walks

Quantum Circuit Model

[Deutsch, Proc. Roy. Soc. Lon. A 425, 73 (1989)]





Prepare states Apply gates Measure

Quantum Circuit Model

+ Foundation of all other models

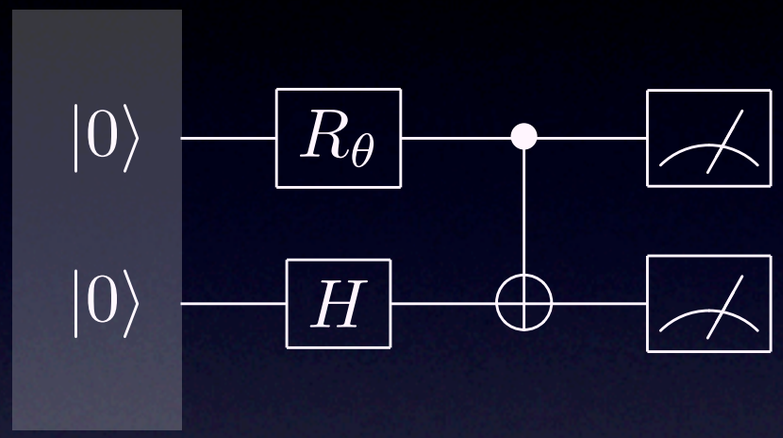
+ Fault-tolerant

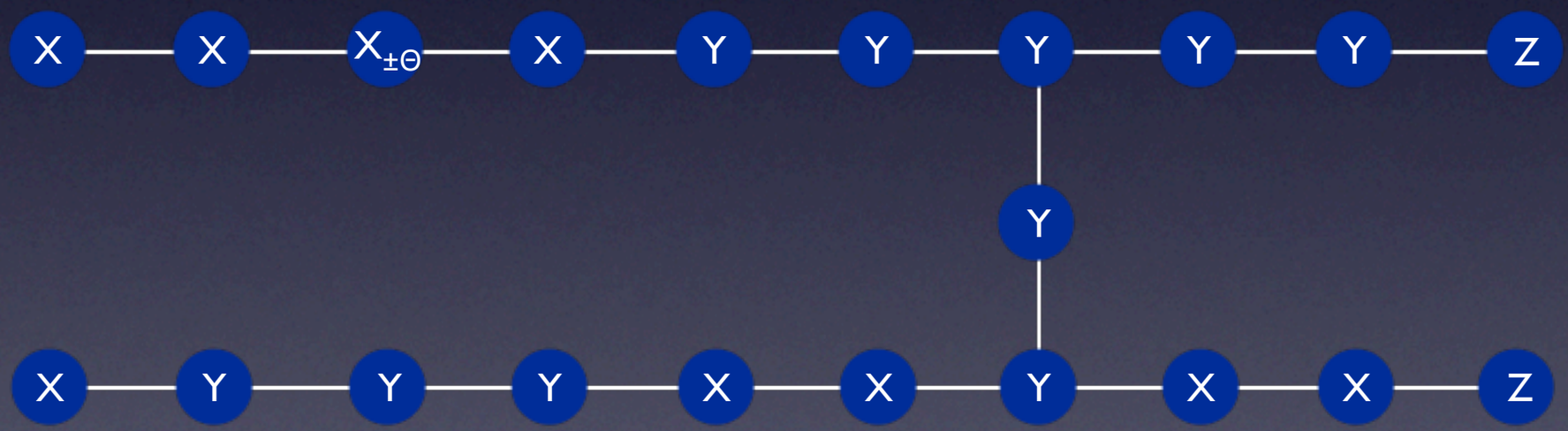
– Unitary gates require precise timing and control

– Separation into three/four procedures each requiring separate hardware that must each individually be optimized

One Way QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]





Adaptive measurement

Cluster State

Entangled State

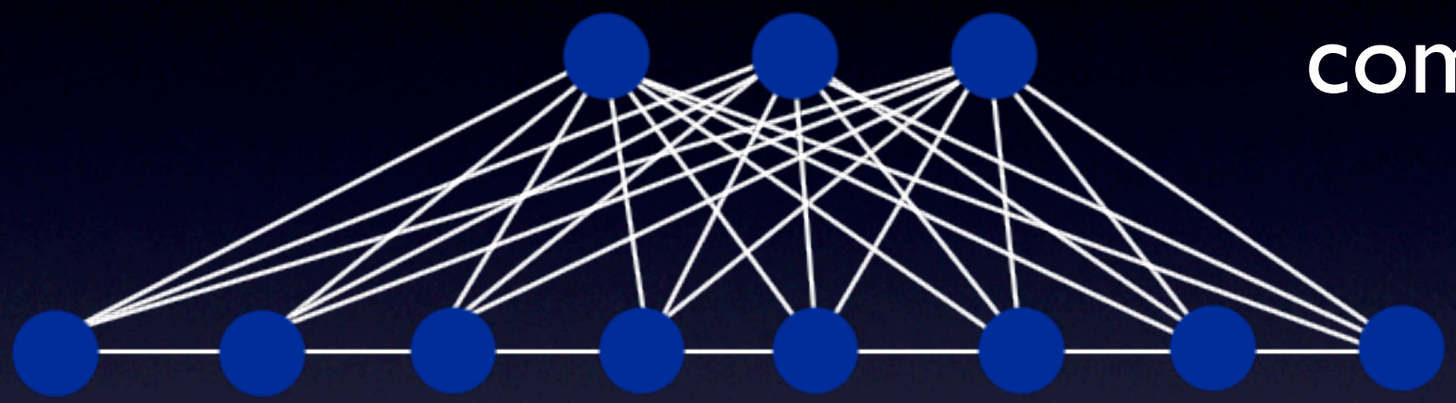
circuit

One Way QC

+ "No" unitary gates
+ Entanglement as "fuel" for quantum algorithm
+ Some compression in certain circuits
+ Fault-tolerant
- Measurements are often slow (~10 to 1000x)

Universal Adiabatic QC

[Aharonov et al. SIAM Journ. of Comp., 37 166 (2007)]



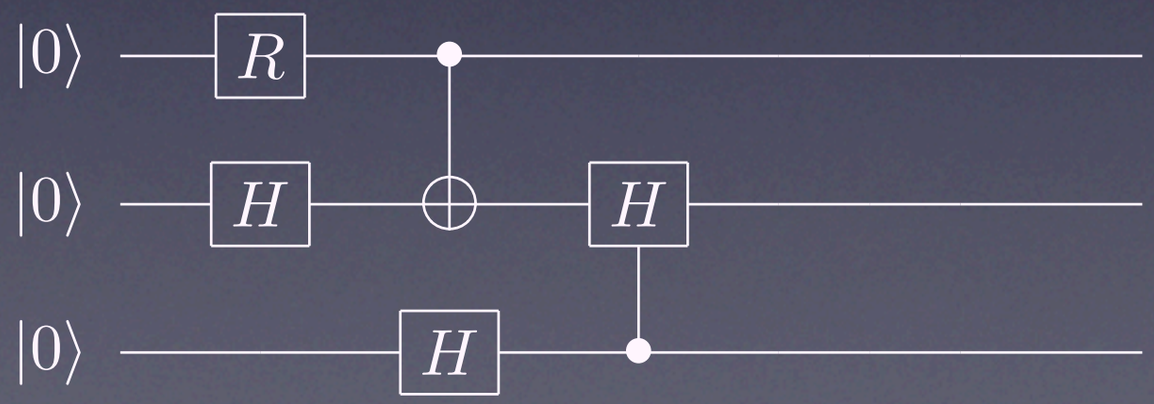
computational qubits

clock qubits

Adiabatically interpolate from H_i to H_f

$$H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$$

Prepare system into ground state



T polynomial in depth of circuit

Final ground state:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle_c V_t |0\rangle$$

$$V_t = \prod_{i=t}^{1} U_i$$

Universal Adiabatic QC

+ Adiabatic gates - fewer “timing” issues

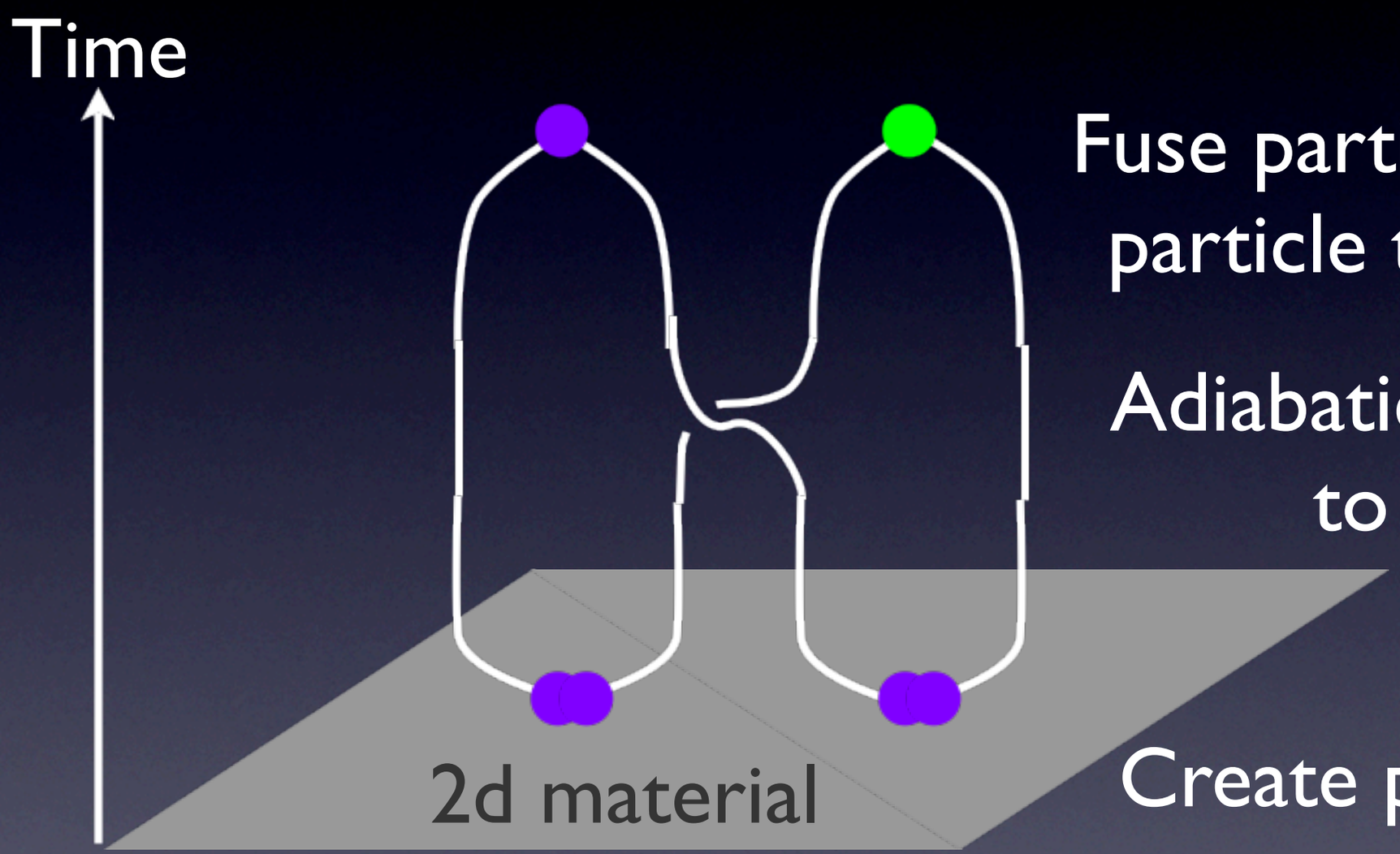
- Not known to be fault-tolerant

- Current constructions use “perturbation theory gadgets” implying small energy gaps even for small circuits

- Uber-complex

Topological QC

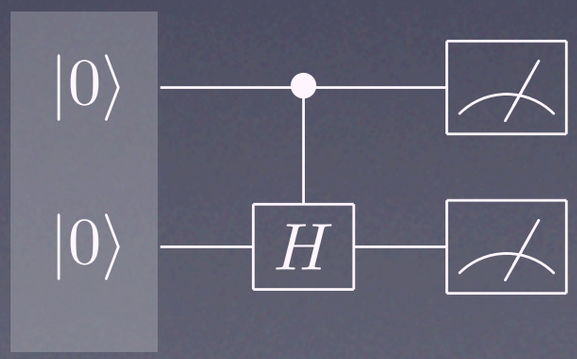
[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]



Fuse particles and observe particle type to measure

Adiabatically braid anyons to enact gates

Create particles (anyons) out of ground state



Topological QC

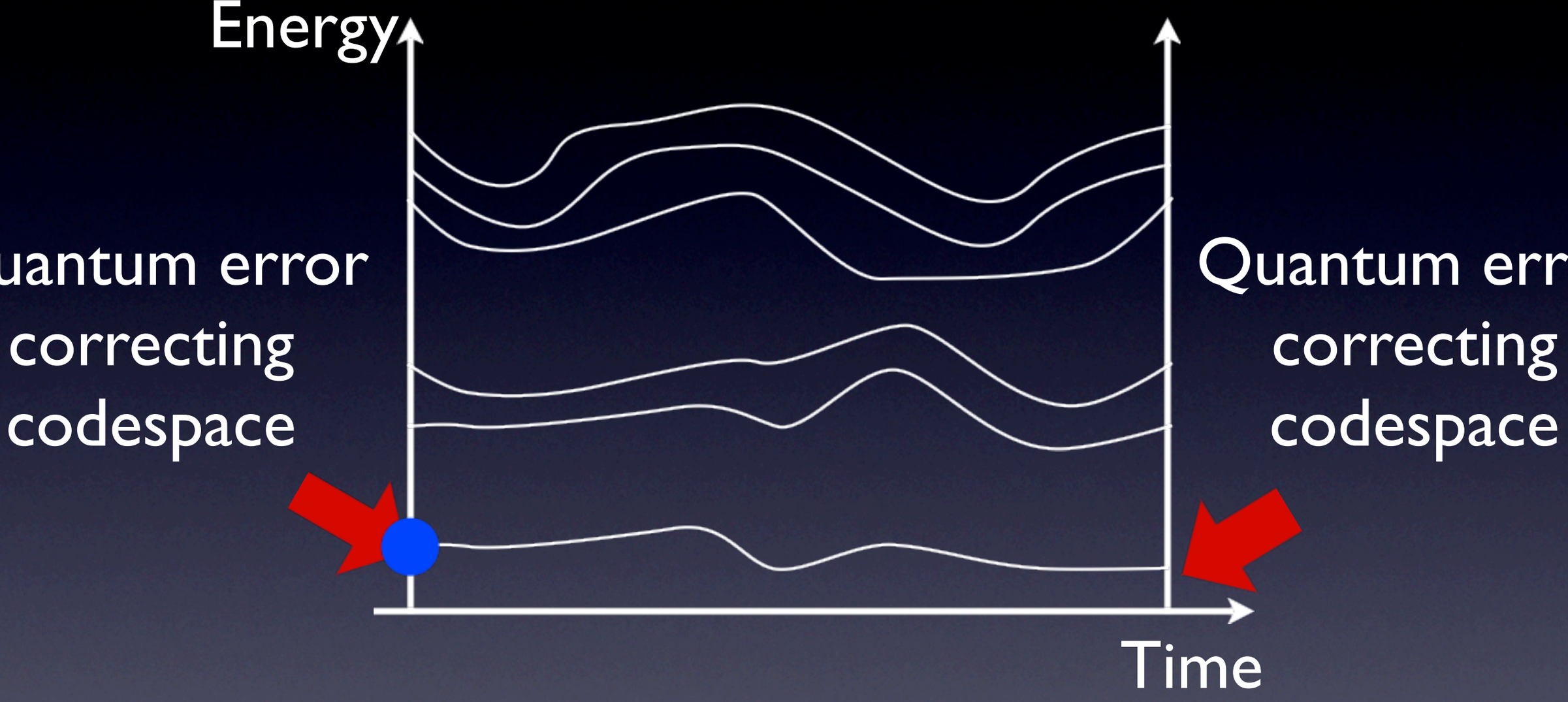
+ Adiabatic gates with topological protection
+ Low decoherence due to energy gap
+ Fault-tolerant?
- No substance (conclusively) known to support non-abelian anyons
- Fault-tolerant?

Today

Three models that combine/reinterpret many of the above models:

- Adiabatic Gate Teleportation
- Adiabatic Cluster State Quantum Computing
- Quantum Transistors

Adiabatic Dragging



Require: degenerate through entire evolution

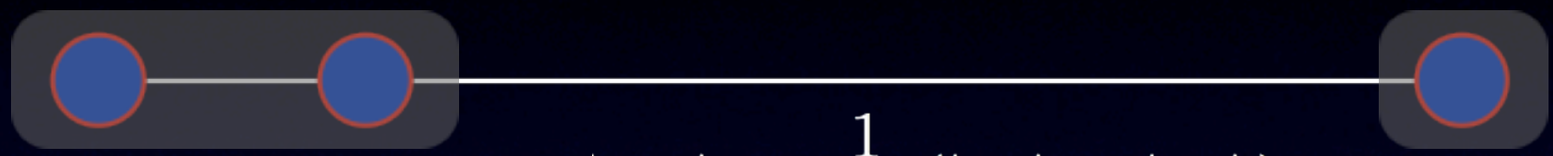
"Open loop" holonomic evolution

[D. Kult, J. Aberg, E. Sjoqvist, Phys. Rev. A 74, 022106 (2006)]

Teleportation

Teleportation Protocol

Alice



Bob

$|\Phi_{\pm}\rangle$ or $|\Psi_{\pm}\rangle$

$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|\psi\rangle$

Measure qubits in Bell basis

Send 2 bits

Conditional on bits apply fixing gate

Initial Code



$|\psi\rangle \otimes |\Phi_+\rangle$

Final Code



$|\eta\rangle \otimes |\psi\rangle$

Adiabatic Teleportation

Initial Code

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Final Code



$|\psi\rangle \otimes |\Phi_+\rangle$



$|\Phi_+\rangle \otimes |\psi'\rangle$

Initial Hamiltonian $\Delta > 0$ Final Hamiltonian

$$H_i = -\Delta(X_2X_3 + Z_2Z_3) \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

Evolution: $H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$

What happens to quantum information in fist qubit?

Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1 X_2 & \bar{Z}_1 &= Z_2 Z_3 \\ \bar{X}_2 &= X_2 X_3 & \bar{Z}_2 &= Z_1 Z_2 \\ \bar{X}_3 &= X_1 X_2 X_3 & \bar{Z}_3 &= Z_1 Z_2 Z_3 \end{aligned}$$

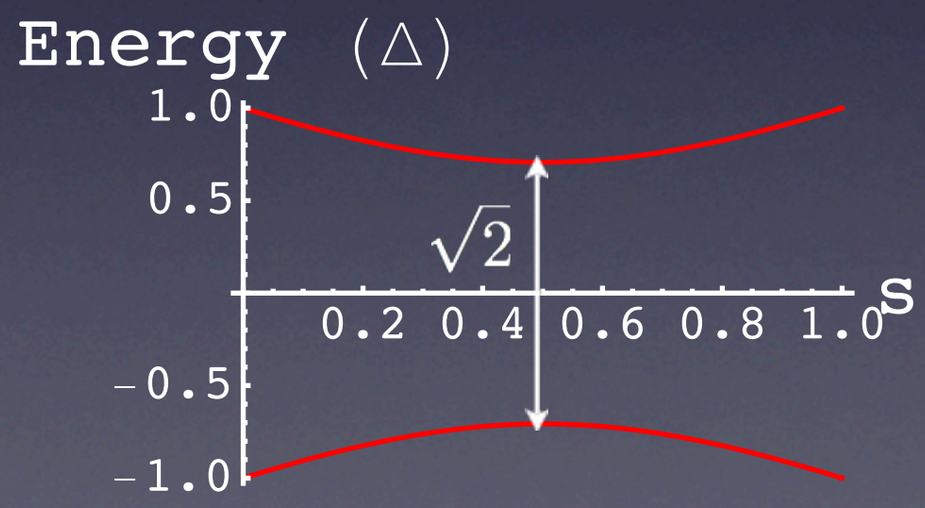
Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

$$H(s) = (1-s)H_i + sH_f$$



Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1X_2 & \bar{Z}_1 &= Z_2Z_3 \\ \bar{X}_2 &= X_2X_3 & \bar{Z}_2 &= Z_1Z_2 \\ \bar{X}_3 &= X_1X_2X_3 & \bar{Z}_3 &= Z_1Z_2Z_3 \end{aligned}$$

Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

Info in first qubit:

$$X_1 = \bar{X}_2\bar{X}_3$$
$$Z_1 = \bar{Z}_1\bar{Z}_3$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

Info in third qubit:

$$X_3 = \bar{X}_1\bar{X}_3$$
$$Z_3 = \bar{Z}_2\bar{Z}_3$$

$$|\psi\rangle \otimes |\Phi_+\rangle \quad \rightarrow \quad |\Phi_+\rangle \otimes |\psi\rangle$$

Adiabatic Gate Teleportation

[Bacon and Flammia, Phys. Rev. Lett. 103, 120504 (2009)]

$$H_i = -U_3\Delta(X_2X_3 + Z_2Z_3)U_3^\dagger \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

Note: spectrum unchanged by unitary

$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes U_3|\psi\rangle$$

Example (Hadamard gate): $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$H_i = -\Delta(X_2Z_3 + Z_2X_3) \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

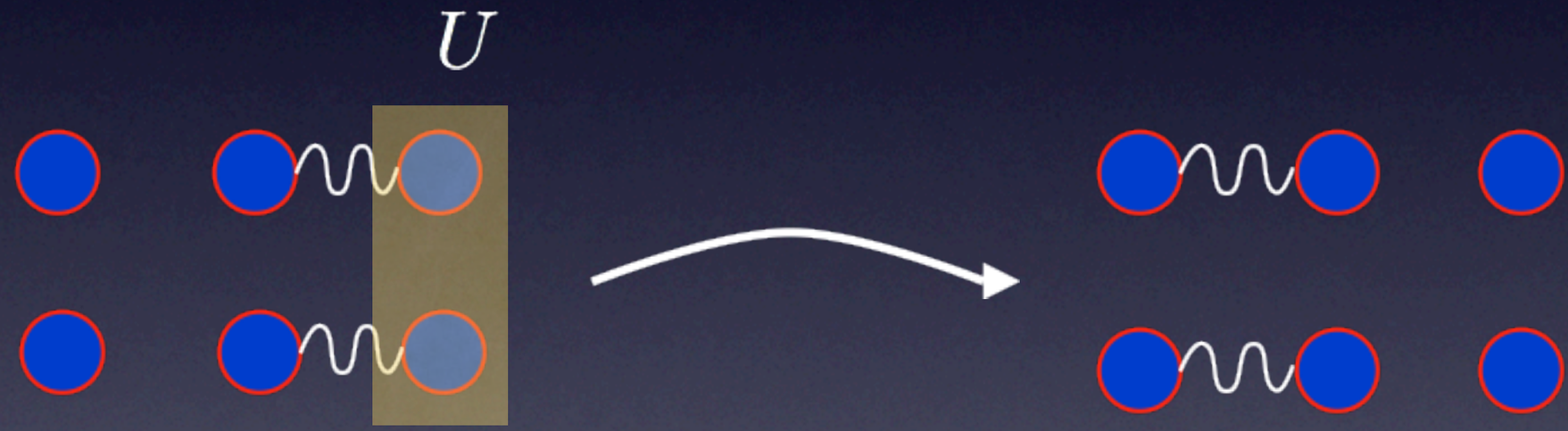
$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes H|\psi\rangle$$

Let's Build a Computer

One qubit gates



Two qubit gates



Problem: initial Hamiltonian requires 3 qubit interactions

Example



$$H_i = -\Delta(X_2X_3Z_6 + Z_2Z_3 + Z_5Z_6 + Z_3X_5X_6)$$

To get 2 qubit interactions: perturbation theory gadgets

Gadgets



Replace 3,6 by encoded qubits: $|0_L\rangle = |00\rangle, |1_L\rangle = |11\rangle$

$$
\begin{aligned}
H_i &= -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a}) \\
&\quad -\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b}) \qquad \lambda \ll \omega \\
H_f &= -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5) \\
&\quad -\omega Z_{3a}Z_{3b}
\end{aligned}
$$

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Gadgets



$$
\begin{aligned}
H_i &= -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a}) \\
&\quad -\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b}) \\
H_f &= -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5) \\
&\quad -\omega Z_{3a}Z_{3b}
\end{aligned}
$$

Minimum gap: $O\left(\frac{\lambda^2}{\omega}\right)$

Initial ground state fidelity: $1 - O\left(\frac{\lambda^2}{\omega^2}\right)$

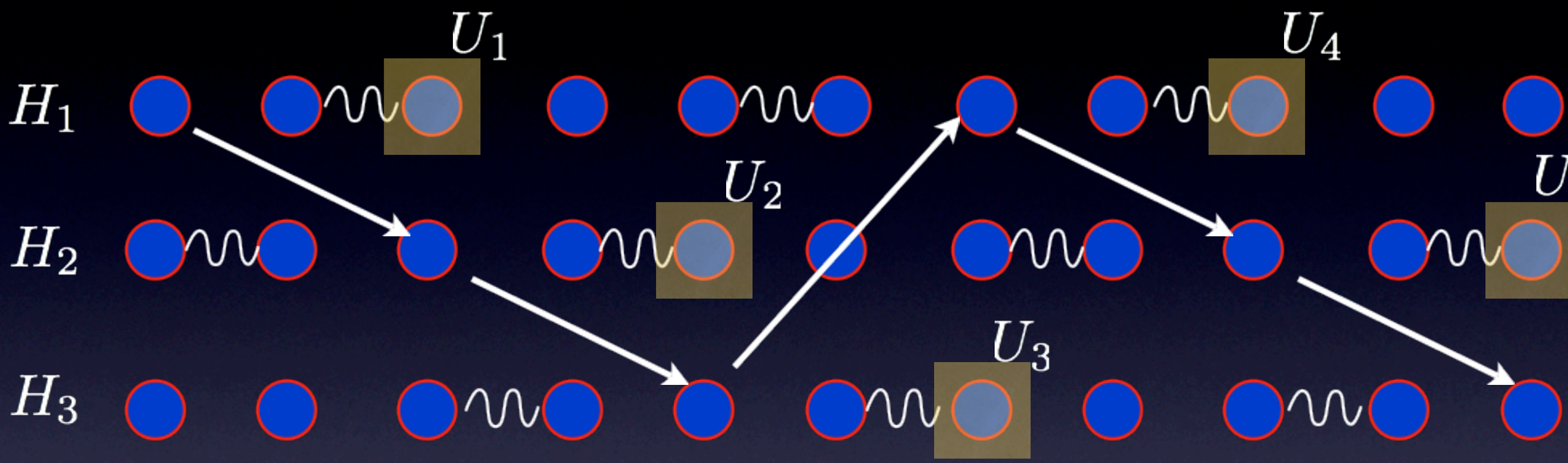
Variations on a Theme

Exchange interactions:

$$H = -\Delta(X_1X_2 + Y_1Y_2 + Z_1Z_2)$$

Architectures





Sequence of Hamiltonians: $H_1, H_2, H_3, H_1, H_2, H_3, \ldots$

Sequence of single qubit gates applied: $U_1, U_2, U_3, U_4, U_5, \ldots$

Generalizing to more than 1 qubit:

Universal QC by cycling between only 3 Hamiltonians

Adiabatic Gate Teleportation

+ Simple adiabatic gates
+ Fault-tolerant (port from gate model)

- Gadget construction slows down 2-qubit gates (only second order, though)
- Still requires state preparation and final measurement

Anything You Can Do I Can Do Adiabatically?

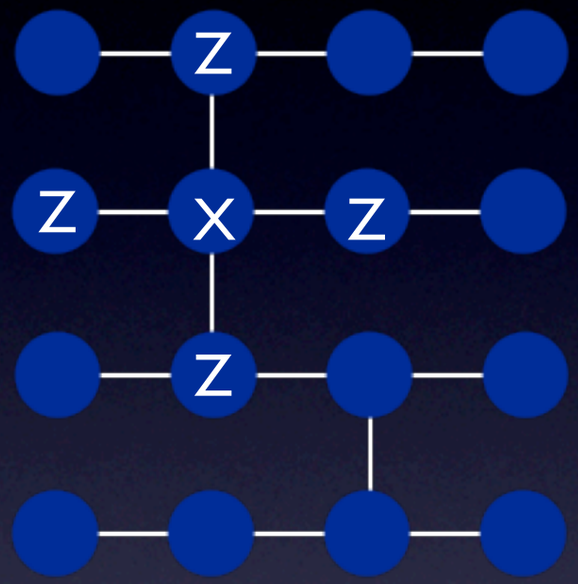
- Adiabatic Gate Teleportation takes a conditional measurement based protocol and performs it adiabatically

- Can other such protocols be similarly converted (like the one-way model?)



"Anything you can do, I can do adiabatically, I can do anything slower than you..."

Cluster State Hamiltonian



$$S_v = X_v \prod_{w \text{ adjacent to } v} Z_w$$

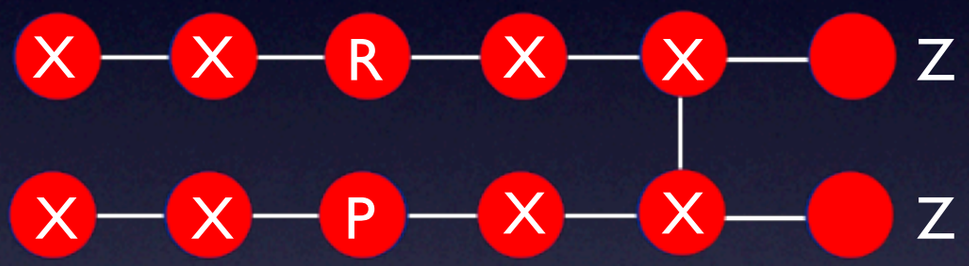
Cluster state is ground state of $H_C = -\Delta \sum_v S_v$

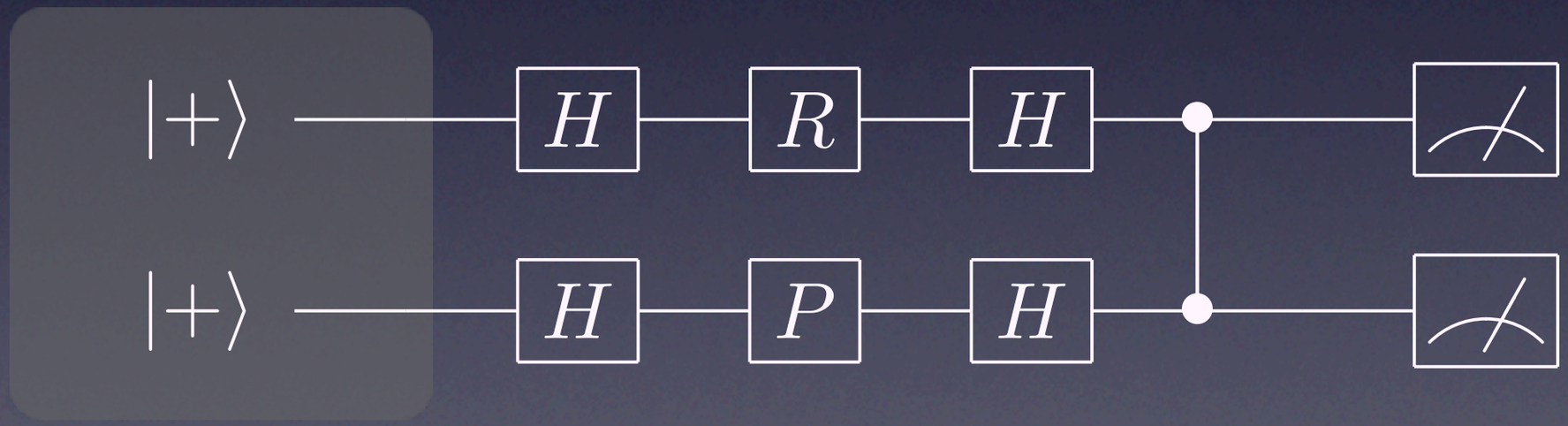
Possible to use gadgets to make only 2-qubit interactions:

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Adiabatic Cluster State QC

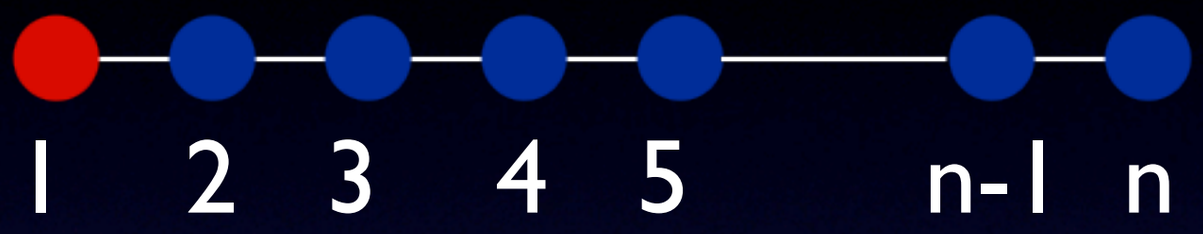
By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:





[Bacon and Flammia, Phys. Rev. A 82, 030303R (2010)]

One Qubit



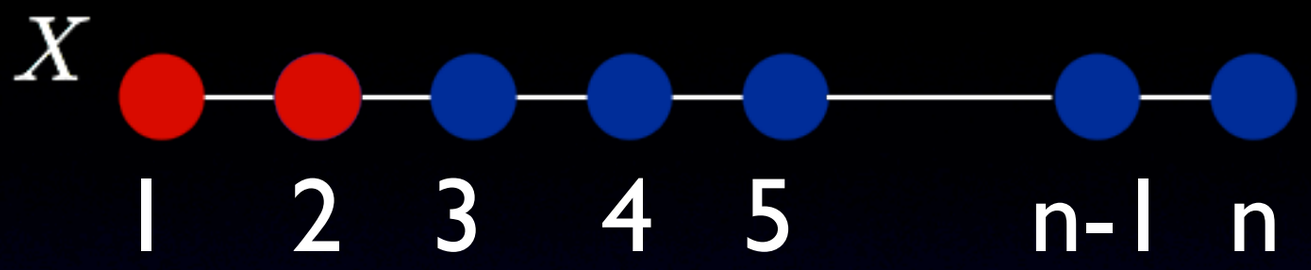
Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$
$$S_{n-1} = [Z]_{n-1}[X]_n$$

Ground state is 2-fold degenerate, can define logical operators for this qubit:

$$\bar{X} = [X]_1[Z]_2$$
$$\bar{Z} = [Z]_1$$

One Qubit



$$S_i = [Z]_i[X]_{i+1}[Z]_{i+2}\ 2 \leq i \leq n-2$$

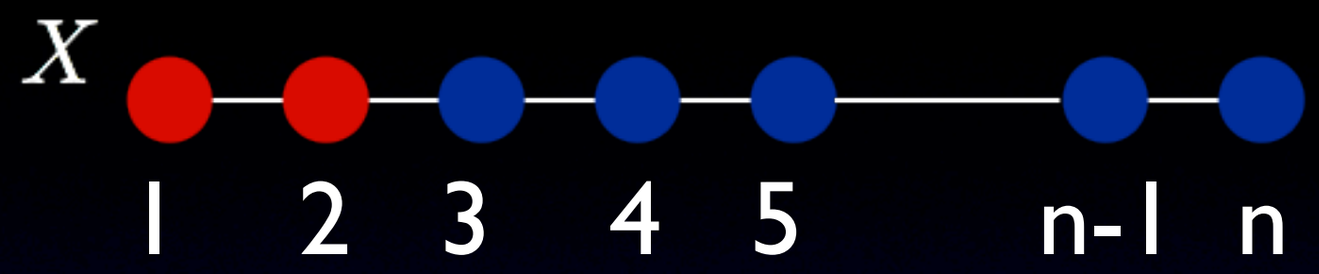
$$S_{n-1} = [Z]_{n-1}[X]_n$$

$$\bar{Z}_1 = [Z]_1[X]_2[Z]_3 \qquad \bar{X}_1 = [X]_1$$

$$\bar{Z} = [X]_1[Z]_2 \qquad \bar{X} = [X]_2[Z]_3$$

$$H_C = -\Delta\bar{Z}_1 - \Delta\sum_{i=2}^{n-1} S_i \quad \rightsquigarrow \quad H'_C = -\Delta\bar{X}_1 - \Delta\sum_{i=2}^{n-1} S_i$$

One Qubit Information



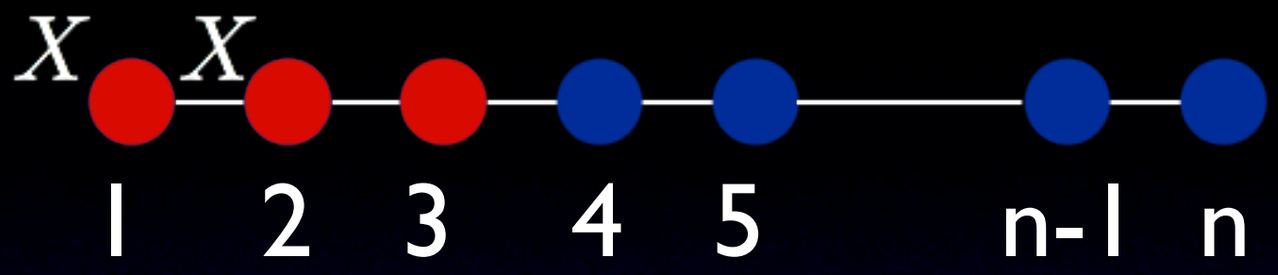
$$H_C = -\Delta \bar{Z}_1 - \Delta \sum_{i=2}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2}\ 2 \leq i \leq n-2$$
$$S_{n-1} = [Z]_{n-1}[X]_n$$

$$H'_C = -\Delta \bar{X}_1 - \Delta \sum_{i=2}^{n-1} S_i \qquad \bar{Z}_1 = [Z]_1[X]_2[Z]_3 \qquad \bar{X}_1 = [X]_1$$
$$\bar{Z} = [X]_1[Z]_2 \qquad \bar{X} = [X]_2[Z]_3$$

$$\bar{X}_{\text{initial}} = [X]_1[Z]_2 = \bar{Z}$$
$$\bar{Z}_{\text{initial}} = [Z]_1 = \bar{X}\bar{Z}_1$$

Information propagates one qubit down the line and H is applied.

State Preparation



$$H_C = -\Delta \sum_{i=0}^{n-1} S_i$$ non-degenerate

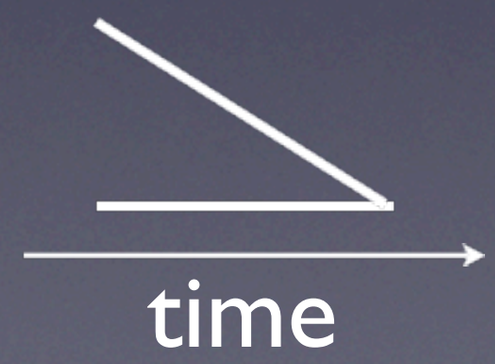
$$S_0 = [X]_1[Z]_2$$
$$S_i = [Z]_i[X]_{i+1}[Z]_{i+2}$$
$$1 \leq i \leq n-2$$
$$S_{n-1} = [Z]_{n-1}[X]_n$$

$$H'_C = -\Delta \sum_{i=3}^{n-1} S_i - \Delta([X]_1 + [X]_2)$$

2-fold degenerate

$S_1 S_3 = [X]_1[X]_3[Z]_4$ commutes with both Hamiltonians

$S_1 S_3 = +1$ begin $S_1 S_3 = +1$ end over this subspace evolution is gapped



Adiabatic Cluster State QC

+ Simple adiabatic gates

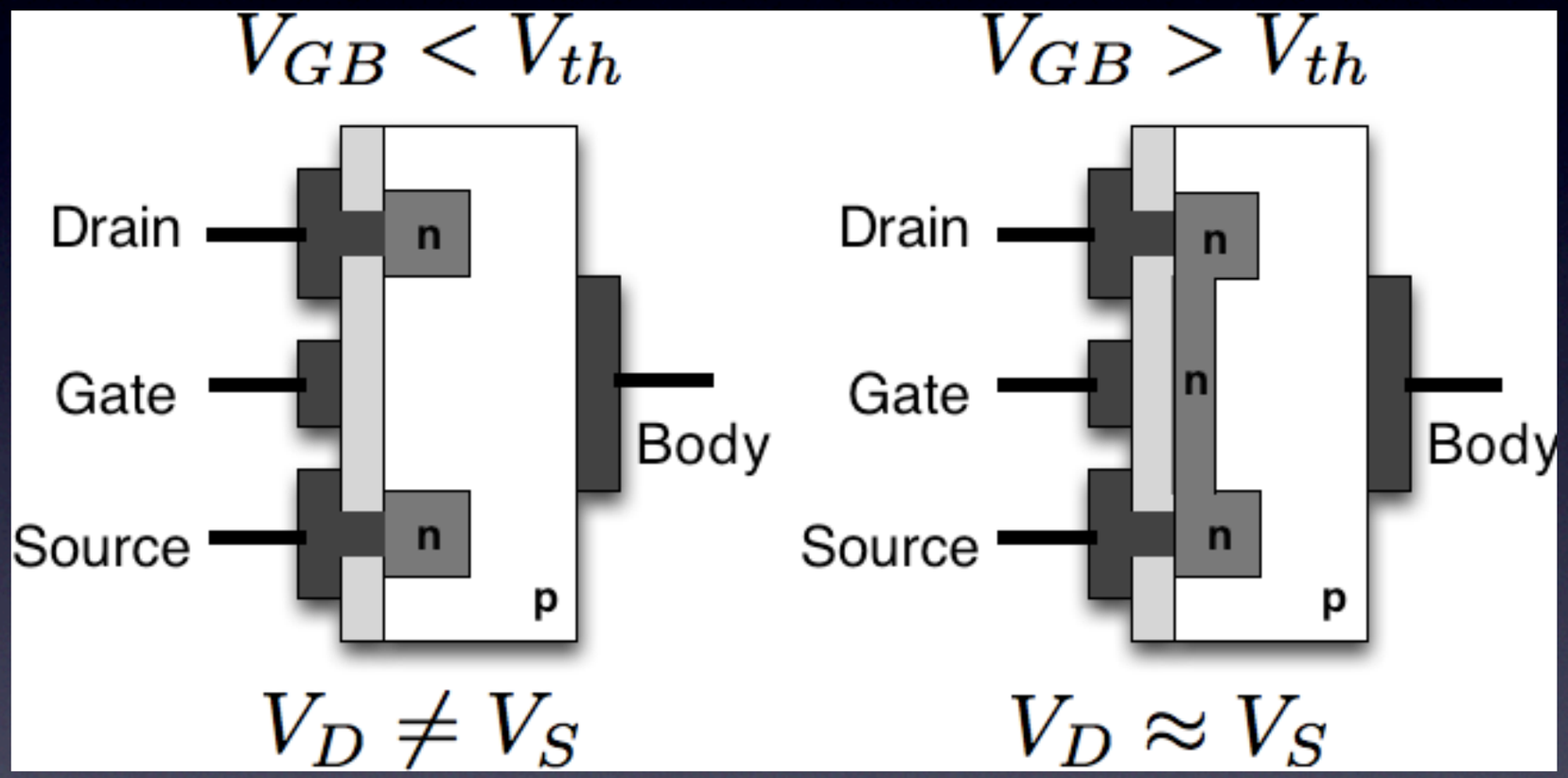
+ Preparation like topological QC, symmetry enforced initialization

+ Requires only final measurement

+ Fault-tolerant

- Gadget construction slows down 2-qubit gates (third order)

Can We Make a Classical Transistor?

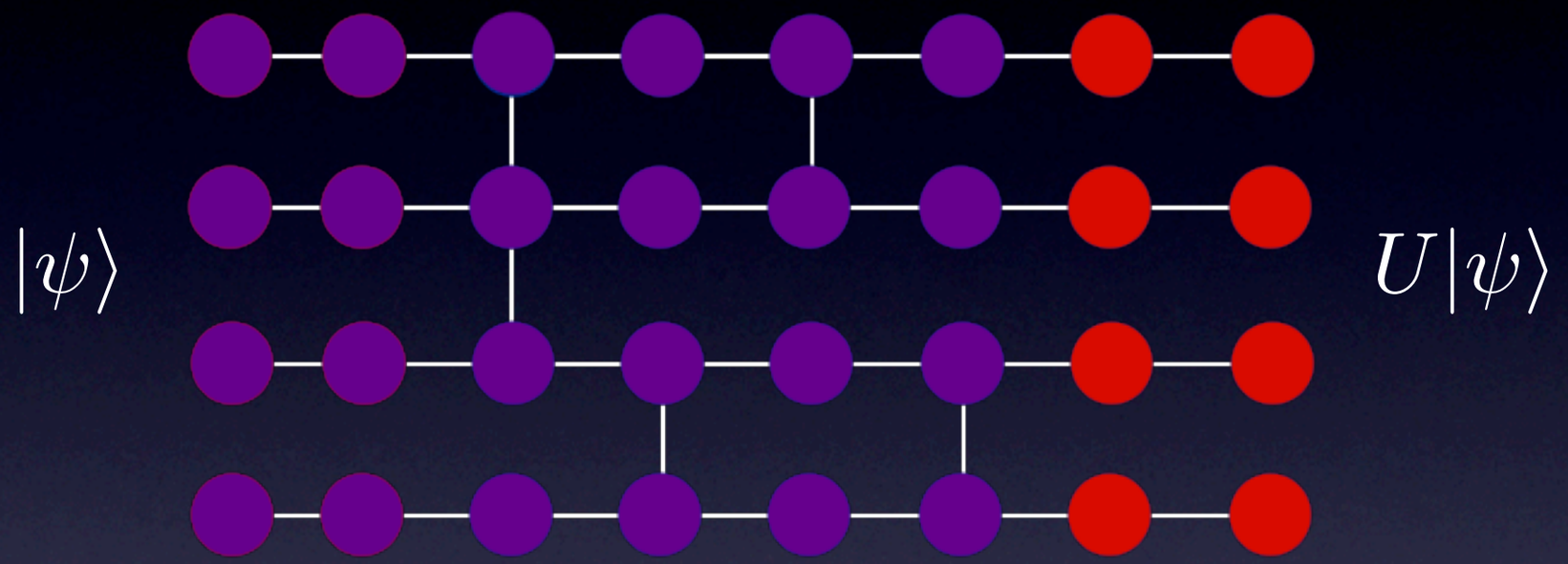


An "identity gate"

Problem: quantum information cannot be cloned

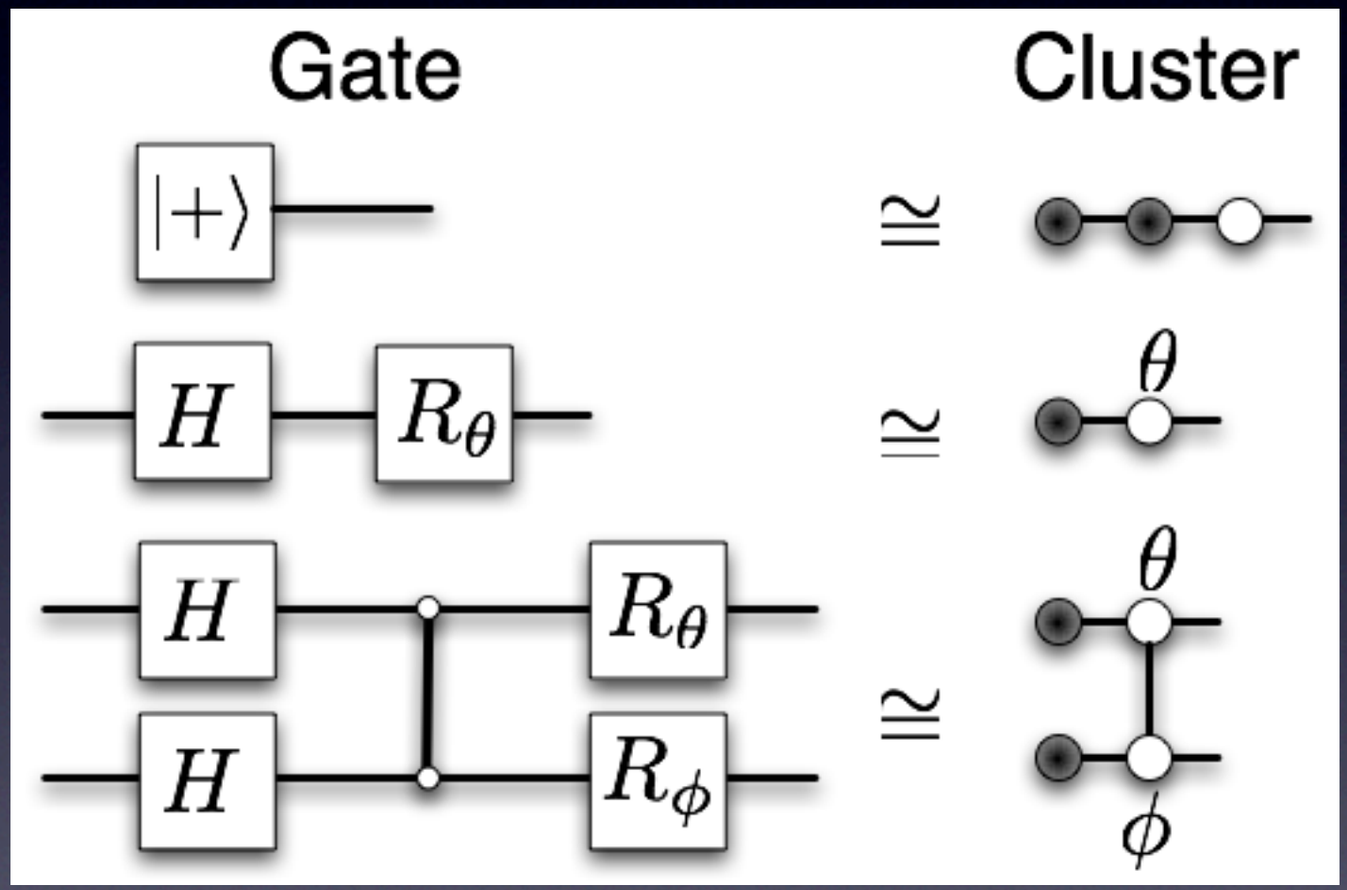
Quantum Transistor

[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]



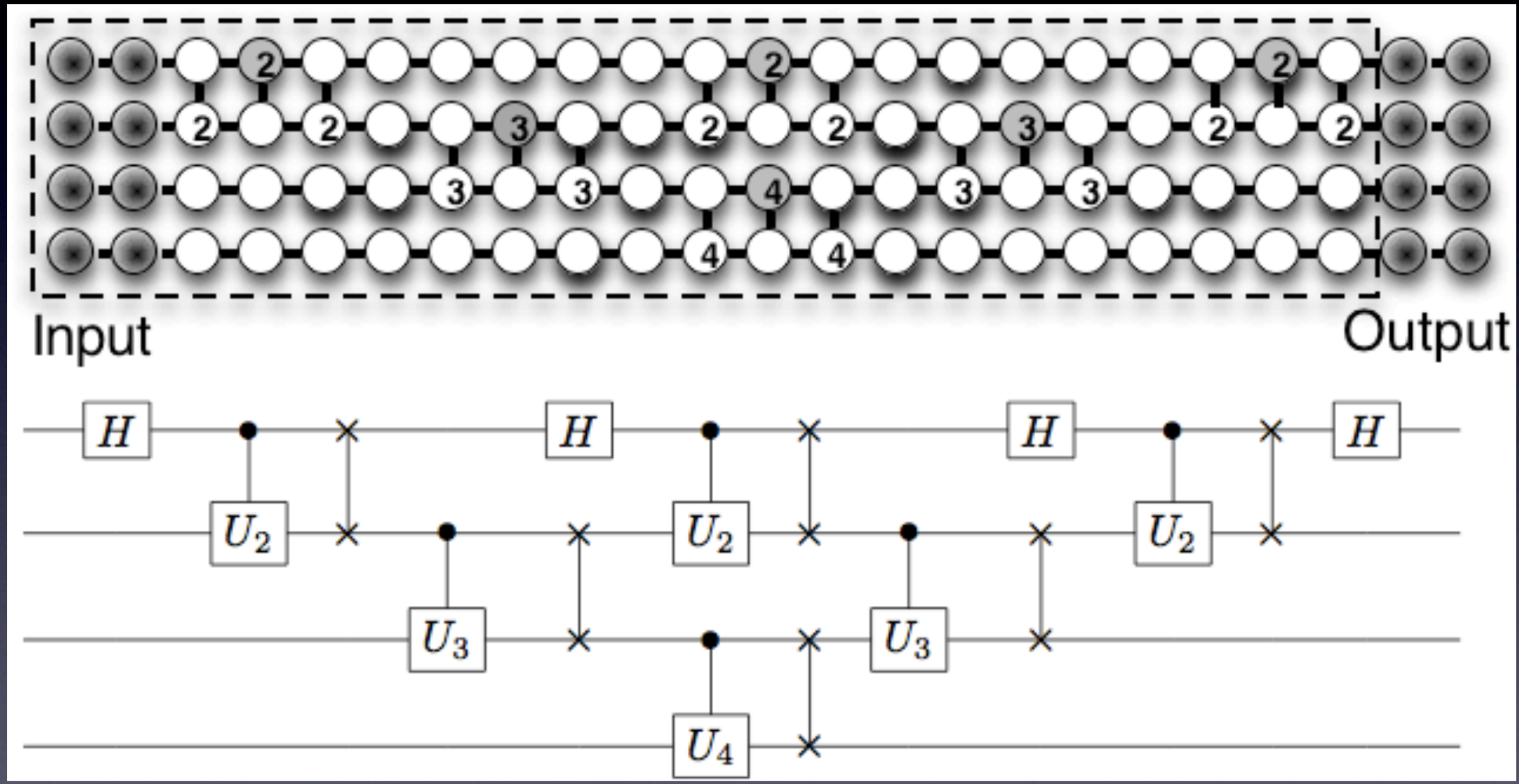
1. Many-body system in its ground state
2. Qubits localized on one side of the device
3. Apply a strong 1-qubit external field to device
4. Qubits now localized on other side of device with a quantum circuit applied to the qubits

Quantum Transistor Dictionary



$$R(\theta) = \exp(-i\theta Z/2)$$

Example

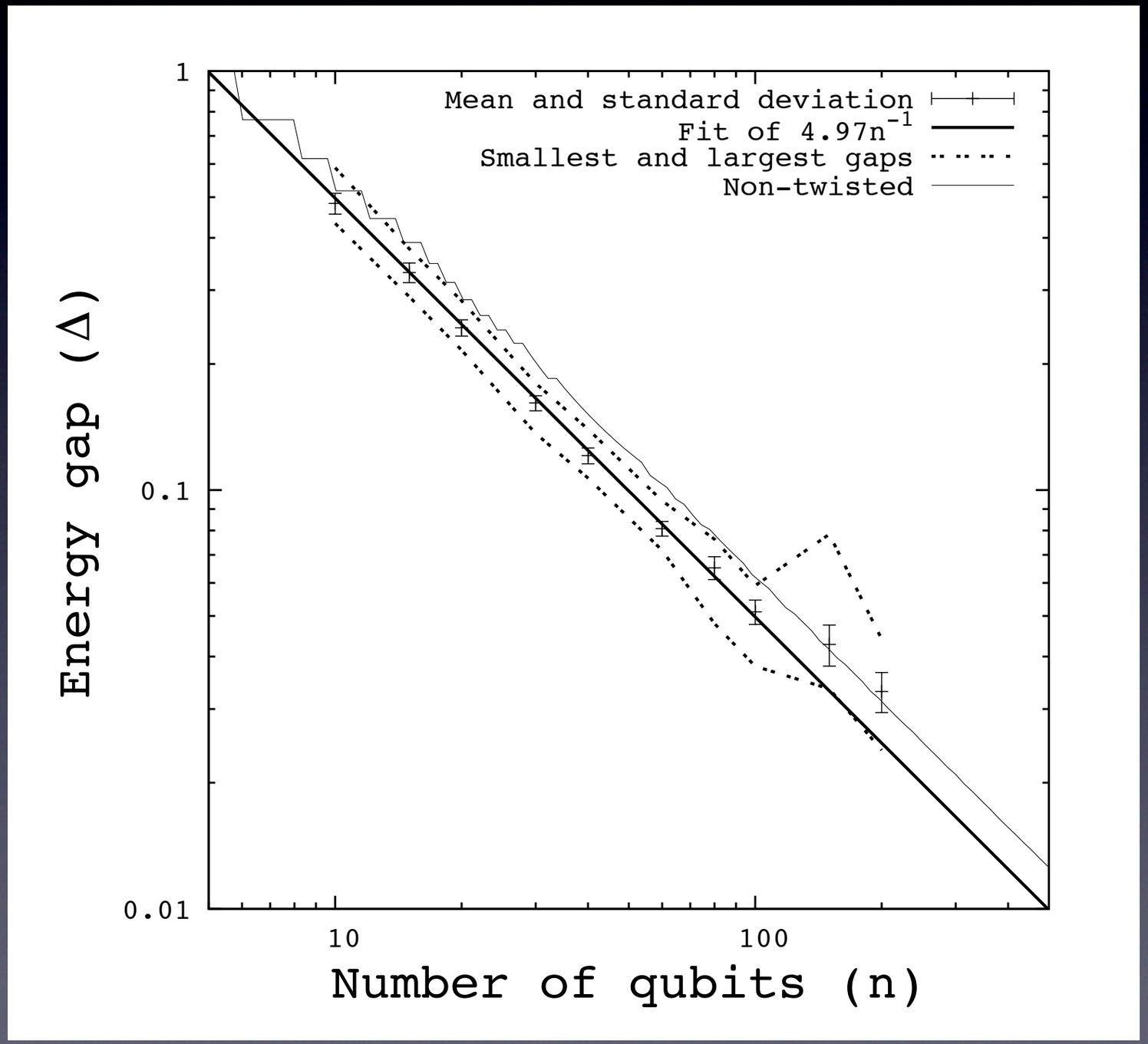


Adiabatic?

- How slowly must we turn on the external field in order for the device to successfully quantum compute?

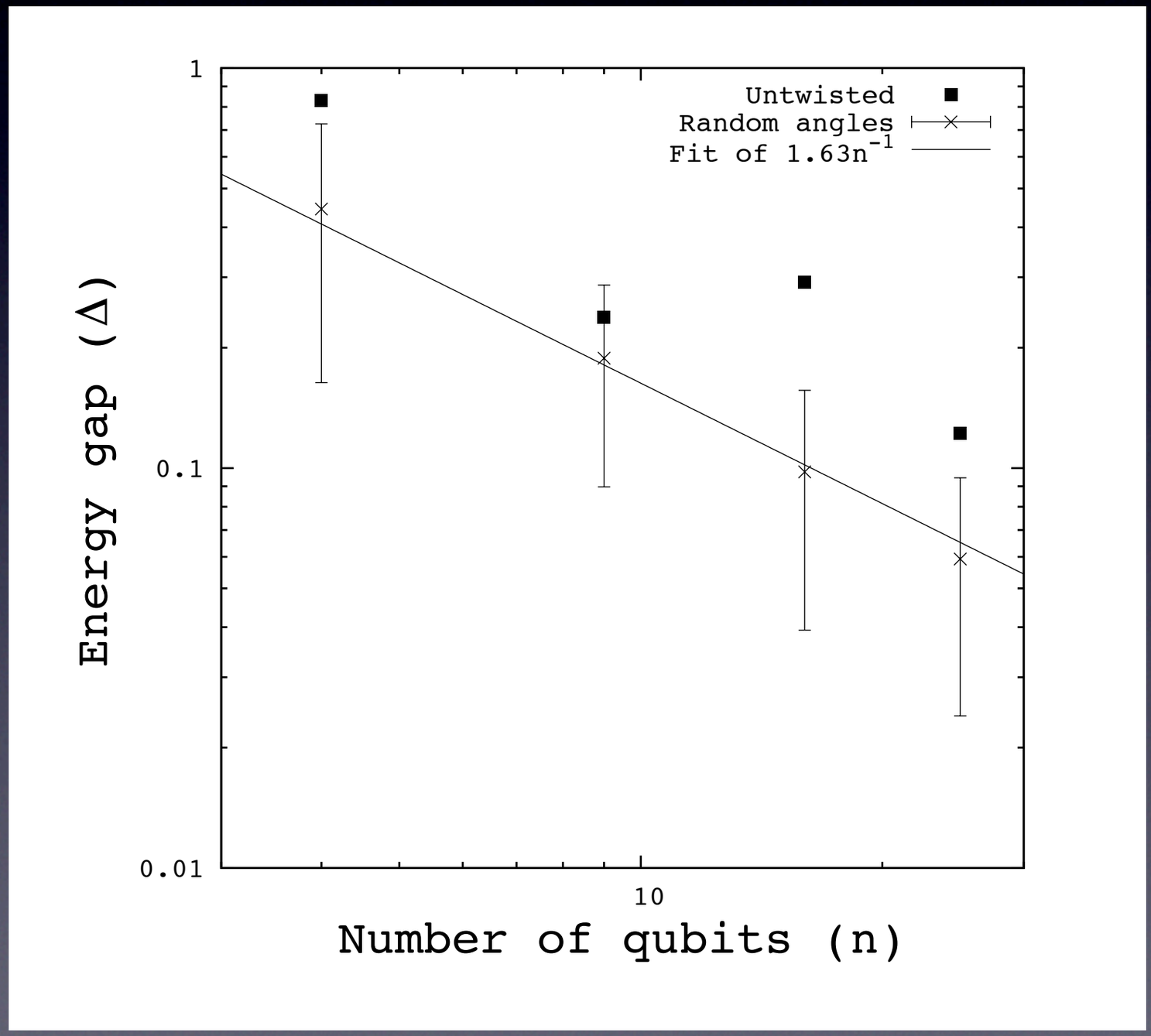
Adiabatic Quantum Transistor

Gap for 1D circuit with random angle rotations...



Adiabatic Quantum Transistor

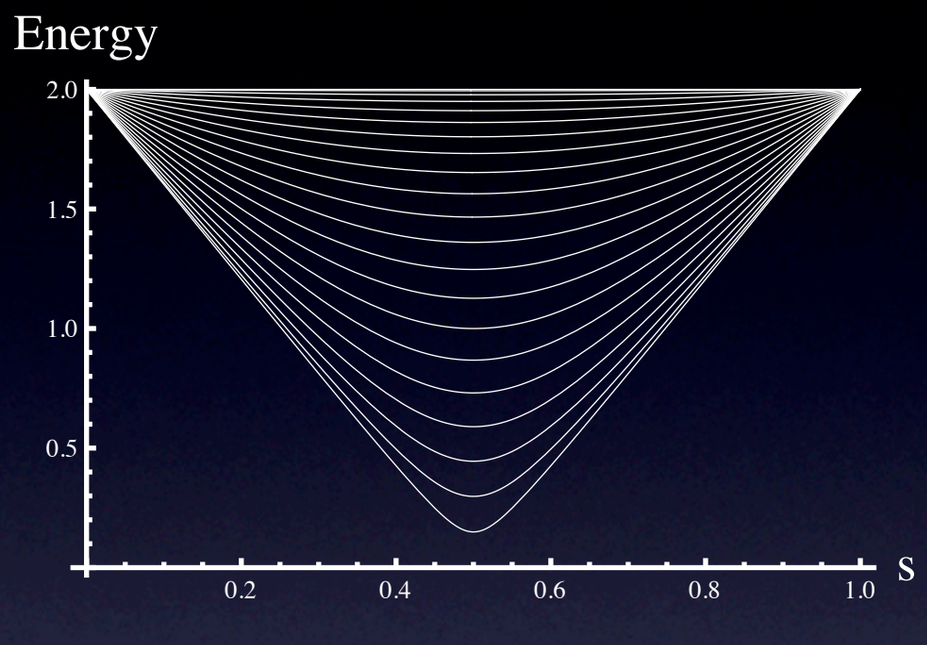
Gap for 2D circuit with random angle rotations...



Adiabatic

- How slowly must we turn on the external field in order for the device to successfully quantum compute?
  - 1D no twists, proven polynomial in size of circuit (map to 2 transverse Ising models with boundary terms.)
  - 1D twists, extremely strong polynomial in size of circuit
  - 2D twists, evidence of polynomial in size of circuit

Fault-Tolerance?



For 1D untwisted Hamiltonian, decouples into 2 1D Ising chains with transverse field.

Two types of errors:

- Those that change energy
- Those that act on degeneracy

Which lead to

- Errors that change energy can be mapped to circuit model errors (independent error model)
- Degeneracy errors only don't cost energy at small and large times. Adjust schedule to make independent

1D errors mapped to independent error model

Quantum Transistors

+ Adiabatic gates, state prep is only in initializing ground state
+ Only need to apply field across portion of device (clocked architecture)
+ Requires only final measurement
+ Universal modular adiabatic quantum computing
- Arguments for fault-tolerance, but not proven

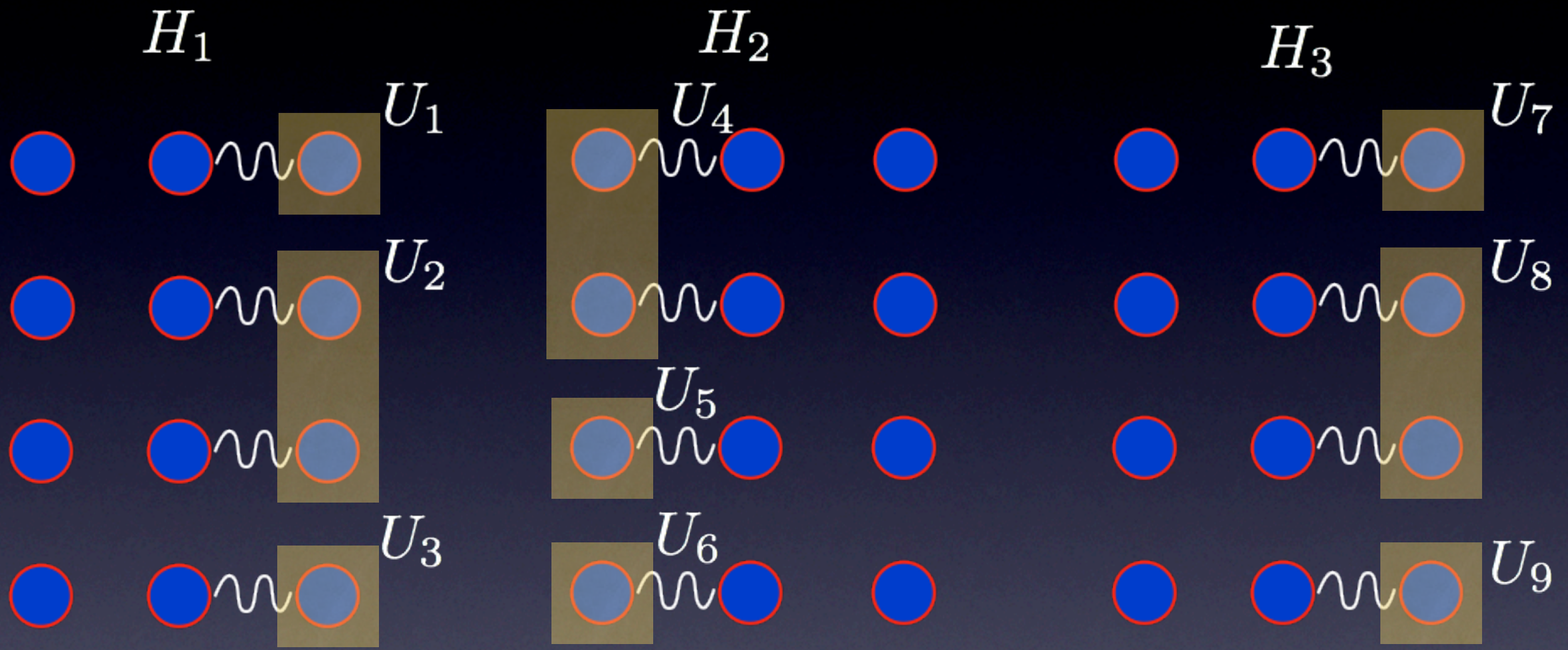
Help! There's a Computer in My Physical System!

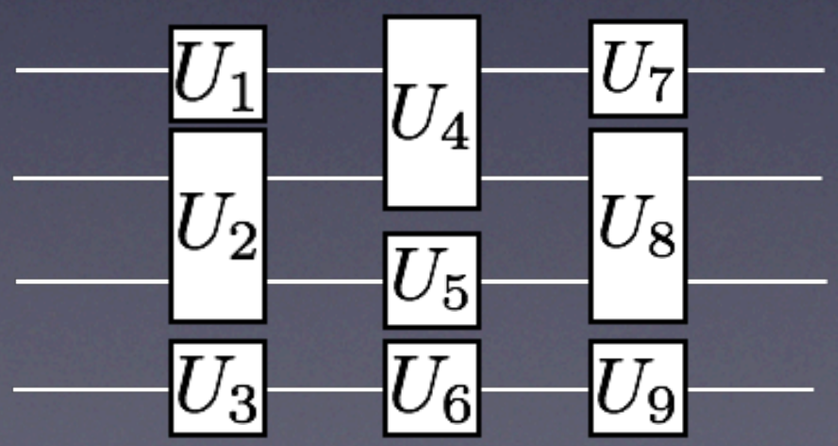


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Minimal Architectures







n qubit circuit

4n or 5n qubits

"Q-Dub" Group

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Jonathan Shi (Physics/CS)

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